UNIVERSITY OF CALIFORNIA,
IRVINE

A Graphical Interface for Specifying and Establishing Multi-Robot Formations

THESIS

submitted in partial satisfaction of the requirements
for the degree of

MASTER OF SCIENCE

in Electrical and Computer Engineering

by

Riwa Karam

Thesis Committee:
Dean Magnus Egerstedt, Chair
Professor Maxim Shcherbakov
Professor Yanning Shen

2025

The work behind this thesis has been submitted as a regular paper to the 2025 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)

DEDICATION

To my family and friends, for their unwavering support and prayers.
And to the robots, for patiently enduring every code revision with no complaints.

TABLE OF CONTENTS

LIST OF FIGURES

LIST OF TABLES

LIST OF ALGORITHMS

ACKNOWLEDGMENTS

Lumos! This thesis has enlightened me to a great love for research and strengthened in me the constant pursuit of learning and understanding.

I would like to express my deep gratitude to Professor Magnus Egerstedt for his unwavering guidance, patience, and insightful feedback throughout this thesis. His expertise and confidence in my work have been a constant source of motivation; his forever optimism and positive look towards life and work is always inspiring.

I would like to thank Professors Maxim Shcherbakov and Yanning Shen for being on my committee and for being amazing teachers and researchers. Their work ethic and love for their craft is something to look up for and learn from.

I also extend sincere thanks to my lab mates Alex Nguyen, Ruoyu Lin, and Brooks Butler for their thoughtful critiques and invaluable discussions. Their contributions helped sharpen my ideas and strengthen the foundation of this thesis. My colleagues in the UCI Robot Ecology Lab played a key role through their support, sharing ideas and offering encouragement at every step.

I am indebted to my best friend Mariam Rakka, who generously provided moral support and a listening ear whenever I encountered a research or personal hurdle. Her faith, positive energy, and hard work were and always will be an example for me, and kept me moving forward.

Finally, I owe my deepest appreciation to my parents Walid Karam and Marie Baz and my sisters Myriam and Farah Karam, for their unwavering belief in my abilities, reminding me that no challenge is insurmountable.

ABSTRACT OF THE THESIS

A Graphical Interface for Specifying and Establishing Multi-Robot Formations

By

Riwa Karam

Master of Science in Electrical and Computer Engineering

University of California, Irvine, 2025

Dean Magnus Egerstedt, Chair

In this thesis, we introduce an easy and user-friendly approach designed to make the specification of robotic swarm formations more accessible through digital drawings provided on a graphical user interface. Traditional approaches often require predefined information such as the exact number of robots and their desired pairwise distances, creating a boundary and limiting usability only to users within the controls and robotics field. To overcome those bounds, our method leverages computer vision techniques to interpret user-drawn sketches, efficiently extracting critical points that define the intended formation structure without needing any other user input. Our framework opens the door for non-expert users to deploy robot swarms, establishing desired robot formation.

The extracted critical points, identified from user sketches, are utilized to systematically construct a rigid graph representation, accurately capturing the intended geometry and spatial relationships of the desired formation. To optimally assign individual robots to their respective positions within this formation, we integrate the Hungarian algorithm, renowned for effectively solving the linear assignment problem. This choice ensures an optimal, cost-effective allocation of robots to formation nodes, maintaining both spatial consistency and deployment efficiency.

Furthermore, recognizing the importance of maintaining the shape of the desired formation,

we conduct a theoretical analysis to ensure its structural integrity. This analysis verifies that the generated formation is free from unintended flips, thus preserving their intended configuration throughout deployment and operation. That analysis and guarantee are necessary since traditional formation control algorithms may allow for rotations, translations, and mirroring of specified formations, to happen; however, our approach requires user-defined formations that can't tolerate potential flipping (i.e., mirroring) with respect to the original user-drawn sketch. Those occurrences take place when the drawings are non-symmetrical.

Experimental and simulation results were conducted using the MATLAB Robotarium simulator and the UCI Robot Ecology Lab Robotarium, demonstrating the practical effectiveness and reliability of our proposed method. Results consistently indicate that formations generated via our approach match user-defined drawings with high accuracy and robustness. Moreover, the scalability and interactive nature of our solution highlight its suitability and adaptability for various complex multi-agent robotic coordination tasks across multiple domains, including search-and-rescue, environmental monitoring, agricultural automation, and collective transportation systems. This thesis, therefore, contributes to advancing human-robot interactions and enhancing multi-agent robotic operational efficiency.

Chapter 1

Introduction

1.1 Motivation

Human-Swarm Interaction (HSI) is an emerging and growing research area dedicated to facilitate human access in specifying, adapting, and deploying robotic swarm behaviors. This field faces numerous technical and operational challenges, particularly focusing on achieving real-time responsiveness, effectively managing distributed swarm dynamics, and balancing the autonomy of robot swarms with centralized human control [1]. As robotic swarms become increasingly deployed in safety-critical and complex environments [2–5], such as disaster response, environmental monitoring, and agricultural management, intuitive and efficient interaction interfaces between human operators and robotic systems have become paramount. The development of such interfaces aims to bridge the inherent complexity of swarm control with user-friendly and accessible systems, enabling even non-expert users to engage meaningfully with sophisticated robotic operations.

Recent research endeavors have significantly emphasized improving the usability and accessibility of HSI systems. Divband et al. [2], for example, presented an interactive, user-centered approach enabling the control of drone swarms by non-specialists. Their approach uniquely integrates the human operator as a part of the swarm itself, improving operational effectiveness in applications like environmental monitoring. Similarly, the incorporation of augmented reality (AR) technologies has emerged as another promising avenue. Sachidanandam et al. [3] demonstrated that AR interfaces significantly enhance human users' capability to guide and manage swarm behaviors, particularly in spatially demanding tasks such as optimal area coverage.

Formation control constitutes one of the fundamental algorithms in swarm robotics, enabling groups of robots to establish and maintain predefined spatial configurations. This capability is critical in various practical scenarios, such as coordinated search-and-rescue operations [6], systematic underwater exploration [7], precision agriculture [8], aerospace formation flights [9], and robotic entertainment displays [10]. However, conventional formation control strategies predominantly rely on predefined parameters, positions, and shapes, thus limiting adaptability and real-time human interaction during mission execution [11–14]. These traditional approaches typically preclude dynamic modifications of swarm configurations in real-time.

Efforts have recently been made to integrate human interaction into formation control algorithms, although they remain constrained by predefined interaction patterns or hand gestures. For example, Suresh et al. [15] developed a gesture-based system, translating predefined hand signals into executable swarm commands. Similarly, Serpiva et al. [4] introduced a neural network-driven gesture recognition system designed to facilitate user-defined swarm trajectories through specific gestures. Despite these advancements, there remains a significant gap in providing truly flexible, intuitive, and unrestricted interaction methods.

Addressing these limitations, our research seeks to fundamentally enhance the accessibility and flexibility of formation control through innovative interaction techniques that support dynamic, real-time, easy, and enjoyable user engagement.

1.2 Contributions

In this thesis, we propose an HSI framework developed for formation control by easily specifying and establishing, on the fly, multi-robotic formations. Our primary objective is for non-expert users to create robotic swarm formations directly through hand-drawn digital sketches without any reliance on predefined gestures or pre-programmed formation structures. This approach extends the capabilities and versatility of human-swarm interfaces, offering freedom in designing swarm behaviors dynamically and interactively.

To achieve this, our method incorporates computer vision techniques to interpret and analyze user-defined digital sketches. These techniques extract critical points that accurately represent the user's intended formation configurations. In turn, the critical points form the foundational structure of a rigid graph, explicitly encoding the spatial and geometric relationships essential for swarm formation control.

Once the rigid graph is constructed, we employ the Hungarian algorithm, a robust and widely recognized method for solving the linear assignment problem, to optimally allocate individual robots to specific formation nodes. This allocation ensures that the formation is realized accurately, and consistently with the user's initial specifications, with a guarantee of having no potential mirroring of the original user input. Additionally, our algorithm explicitly accounts for critical operational constraints, including collision avoidance and spatial integrity, guaranteeing stable formations that do not have any unintended flips or distortions during deployment or operation.

The contributions of this thesis can thus be listed as follows:

1. A user-friendly digital sketch-based interface for HSI, enabling non-expert users to specify and define robotic formations without reliance on predefined or hard-coded formations.

2. An algorithm that leverages computer vision, graph theory, formation control methods, and the Hungarian algorithm for optimal linear assignment, ensuring both efficiency and a no-flip guarantee on the final formation.

3. A detailed theoretical analysis, coupled with experimental validation, establishing the robustness, accuracy, and effectiveness of our method to establish desired formations accurately.

1.3 Organization

This thesis is organized as follows: Chapter 2 provides a comprehensive overview of foundational concepts and methodologies relevant to formation control and graph theory, ensuring a good overview of the preliminaries needed for this work's contributions. Chapter 3 details the proposed no-flip formation algorithm, accompanied by mathematical proofs that substantiate its reliability and stability. Chapter 4 presents experimental and simulation results, demonstrating the effectiveness and reliability of our proposed method. Finally, Chapter 5 synthesizes the contributions of this thesis, discusses broader implications, and outlines future research directions.

Chapter 2

Background

2.1 Formation Control in Swarm Robotics

Formation control is a fundamental approach to multi-agent systems, where multiple agents coordinate and collaborate to achieve and maintain a predefined spatial configuration. By coordinating their movements and exchanging information, locally or globally, agents can efficiently navigate complex environments and complete tasks that are challenging or impossible for a single agent to accomplish alone [16]. The primary objective of formation control is to design strategies that guide agents to their designated relative positions while maintaining desired geometric shapes. This involves addressing challenges such as communication constraints, collision avoidance, dynamic environmental conditions, and input disturbances.

Classic formation control methodologies include:

- **Leader–Follower Strategies:** One or more agents act as leaders that follow preplanned trajectories, and the remaining followers adjust their motions to keep a desired offset from the leader(s) [14]. While simple to implement, this approach can suffer if leader communication is lost.

- **Virtual Structure Approaches:** The entire formation is treated as a single rigid body moving through space; each agent tracks a corresponding point in the virtual structure [17,18]. This yields strong rigidity guarantees but requires global information.

- **Behavior-Based Models:** Inspired by biological swarms, agents follow simple local rules (e.g., cohesion, separation, alignment) to collectively form patterns without explicit distance constraints [19]. These methods are robust to failures but offer weaker formation precision.

Stability and convergence of these methods are typically analyzed via Lyapunov functions or input-to-state stability arguments. For instance, a common potential-based control law for undirected formations can be shown to converge almost globally under connectedness assumptions [20]. Similarly, gradient-based distance-error control laws have been proven to achieve almost-global asymptotic convergence by constructing an appropriate Lyapunov potential function and leveraging graph connectivity properties [21].

2.1.1 Formation Control using Graph Theory

Graph theory provides an appropriate framework for modeling and analyzing interactions among agents in formation control systems. A formation's agents and their respective inter-agent distance constraints can be abstracted as a graph $G = (V, E)$, where $V = \{v_1, v_2, \ldots, v_n\}$ is the set of nodes representing agents, and $E \subseteq V \times V$ is the set of edges representing desired distance constraints [22].

A graph G is *rigid* if the relative positions of its nodes $p_i \in \mathbb{R}^d$ cannot change without violating at least one edge distance, up to global translations and rotations. Formally:

$$\|p_i - p_j\| = d_{ij}, \quad \forall (v_i, v_j) \in E \tag{2.1}$$

where d_{ij} is the target distance. In 2D, a necessary condition for generic (infinitesimal) rigidity is having at least $2n-3$ edges, and the rigidity matrix must have rank $2n-3$ [11].

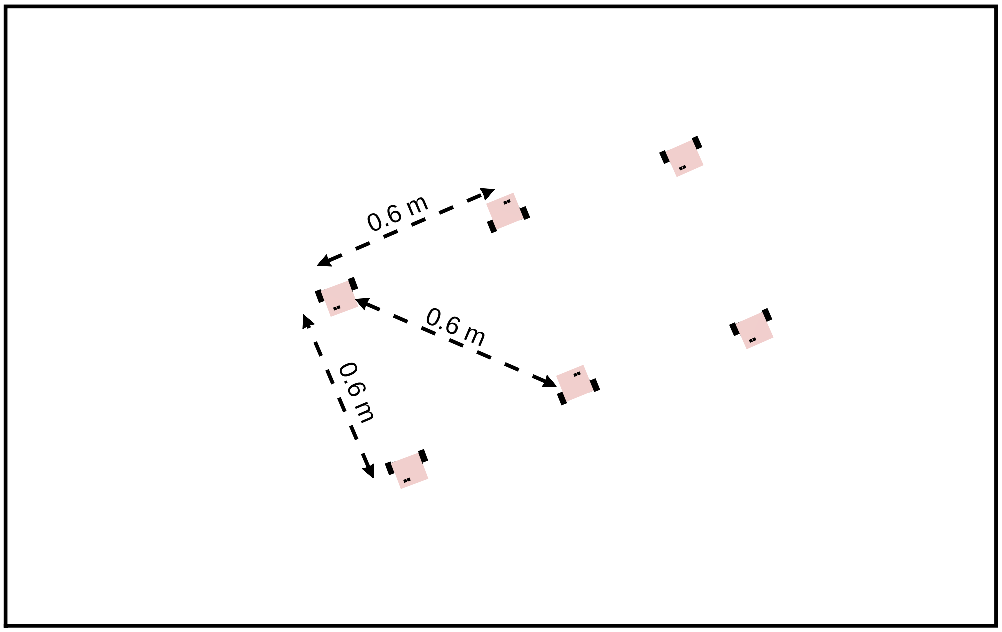


Figure 2.1: Rectangular formation with desired inter-agent distances of 0.6m. The hard-coded Laplacian matrix L determines inter-agent connectivity.

Edges encode the desired distance constraints, and nodes encode agent positions $p_i \in \mathbb{R}^2$. The adjacency matrix A, degree matrix D, and Laplacian $L = D - A$ succinctly capture the interaction topology. A typical decentralized gradient-based control law is:

$$\dot{p}_i = -k \sum_{j \in \mathcal{N}_i} \left(\|p_i - p_j\|^2 - d_{ij}^2 \right) (p_i - p_j), \tag{2.2}$$

where $\mathcal{N}_i$ is the neighbor set of agent i and $k > 0$ is a gain. Under connectivity and rigidity assumptions, this law provably drives the formation error to zero [13]. More specifically, this update rule ensures that agents adjust their positions to satisfy the distance constraints defined by the desired formation. The term $\|p_i - p_j\|^2 - d_{ij}^2$ represents the distance error between connected agents, and the factor $(p_i - p_j)$ guides the correction direction to reduce this error. Through this formulation, agents converge to positions that maintain the desired formation structure while adhering to graph-theoretic constraints. Figure 2.1 shows us a simple example of how formation control works given a pre-defined Laplacian matrix and a desired inter-agent distances of 0.6m. The Laplacian matrix of such a formation would be as follows:

$$L = \begin{bmatrix} 3 & -1 & 0 & -1 & 0 & -1 \\ -1 & 3 & -1 & 0 & -1 & 0 \\ 0 & -1 & 3 & -1 & 0 & -1 \\ -1 & 0 & -1 & 3 & -1 & 0 \\ 0 & -1 & 0 & -1 & 3 & -1 \\ -1 & 0 & -1 & 0 & -1 & 3 \end{bmatrix} \tag{2.3}$$

Practical extensions consider directed graphs (for one-way communication) [23], time-varying topologies (to handle link failures) [24], and additional collision avoidance or obstacle avoidance potential fields [25].

2.2 Computer Vision

Computer vision (CV) is a multidisciplinary field that seeks to endow machines with the ability to interpret and understand visual data from images or video streams. Core tasks in CV include image classification, object detection, semantic and instance segmentation, depth estimation, optical flow, and visual simultaneous localization and mapping (SLAM) [26].

At the heart of most modern CV systems, lies a processing pipeline, from which we note:

Table 2.1: Stages of a computer vision pipeline

Stage	Description
Pre-processing	Raw images are pre-processed to reduce noise and normalize intensities. Common techniques include Gaussian filtering, histogram equalization, and color-space transformations. [26, 27]
Edge Detection	Edges are detected using operators such as Canny [28], Sobel [27], or Laplacian of Gaussian [29], to identify intensity discontinuities corresponding to object boundaries.
Point Extraction	Interest points (corners or keypoints) are localized using detectors like Harris–Stephens [30], Shi–Tomasi [31], FAST [32], and SIFT [33], providing salient locations for downstream tasks.

Overall, the stages above transform raw image inputs into crisp geometric cues—cleaned contours and repeatable landmark points that can be directly interpreted as nodes and connections in a formation-specification graph. By abstracting free-hand sketches or visual observations into an explicit set of edges and vertices, these methods provide the perceptual foundation upon which assignment and control strategies build. In this way, reliable detection of boundaries and interest points bridges low-level image data and the higher-level decision processes needed for coordinated multi-agent behaviors.

2.3 Hungarian Algorithm for the Linear Assignment Problem

The Linear Assignment Problem (LAP) seeks a minimum-cost perfect matching in a weighted bipartite graph. Given a cost matrix $C = [c_{ij}]$ for assigning "workers" to "jobs", the objective is formulated as follow:

$$\min_{X \in \{0,1\}^{n \times n}} \quad \sum_{i=1}^{n} \sum_{j=1}^{n} c_{ij} x_{ij} \quad \text{s.t.} \quad \sum_{j=1}^{n} x_{ij} = 1, \quad \sum_{i=1}^{n} x_{ij} = 1 \tag{2.4}$$

The cost c_{ij} then represents the effort required for worker i to complete job j. The binary decision variable $x_{ij} = 1$ indicates that worker i is assigned to job j, and the constraints ensure that each worker takes exactly one job and each job is taken by exactly one worker. Solving this matching problem yields the assignment that minimizes the total cost of all worker–job pairings.

The Hungarian algorithm (originally by Kuhn [34], refined by Munkres [35]) solves the LAP in $O(n^3)$ time via the following key steps:

1. *Row and Column Reduction:* Subtract from each row its minimum element, then do the same for each column. This introduces zeros corresponding to potential assignments.

2. *Zero Covering:* Cover all zeros in the matrix using a minimum number of horizontal and vertical lines. If the number of covering lines equals n, an optimal assignment can

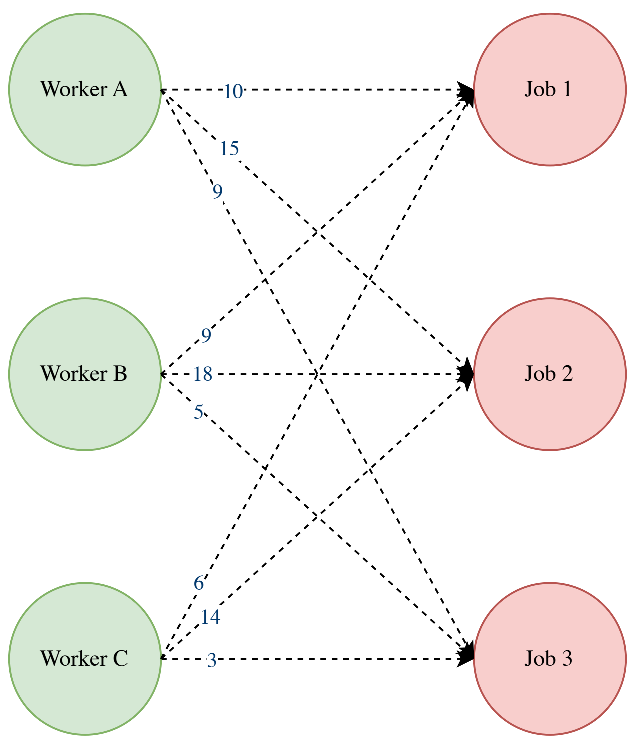


Figure 2.2: Example of a Bipartite Graph with Workers and Jobs. Using the Hungarian Algorithm, each worker on the left-hand side will be assigned to a job from the right-hand side, while minimizing the total cost.

be found among the zeros.

3. *Matrix Adjustment:* Otherwise, find the smallest uncovered element, subtract it from all uncovered elements, and add it to elements covered twice. Repeat the covering step.

4. *Assignment Extraction:* Once n lines cover all zeros, select a set of independent zeros (no two in the same row or column) to form the optimal assignment.

Figure 2.2 shows an example where the Hungarian algorithm could be applied to find the optimal allocation of jobs to workers, by minimizing the total cost.

Chapter 3

No-“Flip” Formation Algorithm

This section presents a new HSI-centered approach to formation control algorithm relying on non-expert user input and leveraging computer vision techniques to interpret user-drawn images, extract critical points, and construct rigid graphs. These graphs are then utilized in the formation control algorithm by computing and using their Laplacian matrices as input. A linear assignment based on the Hungarian algorithm ensures a correct mapping of formation nodes to robots, preventing unwanted flipping of formations (hence, initial sketches); however, there is potential rotation and translation, which is inherent in graph-theoretic formation control. We define a flip as follows:

Definition 3.1. *A flip of a formation occurs when a formation is either mirrored horizontally or vertically due to robot assignments crossing paths, resulting in misalignment.*

Figure 3.1 shows an example of how flipping, as described in Definition 3.1, can affect the formation of an initial input image, thereby leading to an undesired formation.

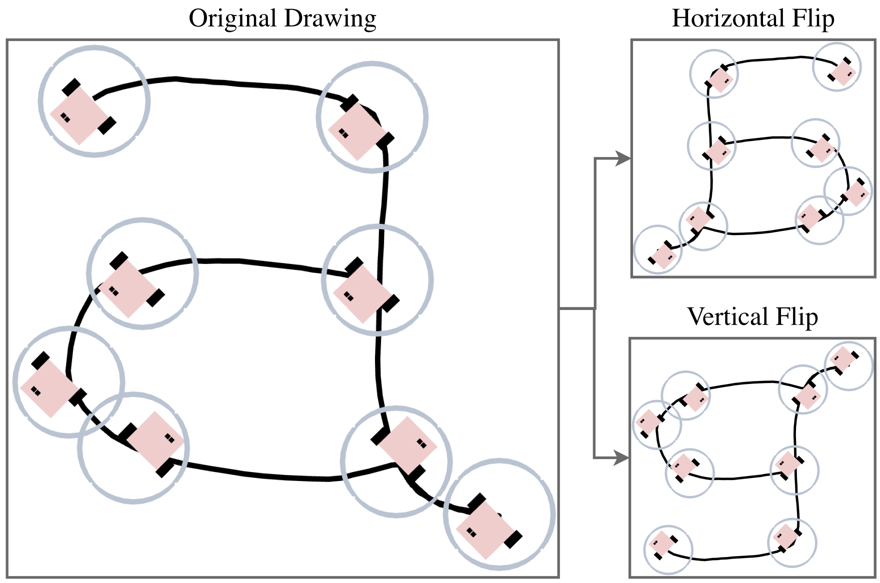


Figure 3.1: Vertical and horizontal flip of the letter "a" as an input drawing.

3.1 Computer Vision

In this section, we discuss the choices of methods used in the computer vision part of the algorithm. Figure 3.2 shows an example of this process with the input user-drawing being a house. There are multiple methods for each step in computer vision processes, and the choice is sometimes difficult to make as it is a combination of multiple consecutive methods. Oftentimes, having understood the best use-case of each method is essential as well as multiple trial and errors to refine the use-case in question. Here is the list of methods used for the purpose of this algorithm:

1. *Image Pre-processing:* The input image is converted to grayscale. This is usually the first step in Computer Vision feature extraction processes; however, for the purpose of this algorithm, this step is redundant since the user-drawn input image is in black (sketch) and white (canvas background color), since colors here are not used in formation control.

2. *Edge Detection:* The Laplacian of Gaussian (LoG) [29] method is employed to iden-

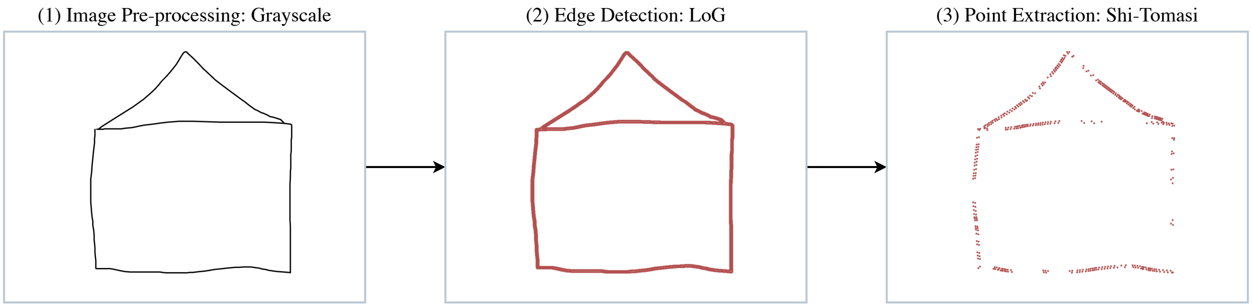


Figure 3.2: Computer vision process example: house as input drawing.

tify edges in the image; it outputs image boundaries as continuous edges. The LoG combines Gaussian smoothing and the Laplacian operator to detect regions with rapid intensity changes.

3. *Point Extraction:* The Shi-Tomasi corner detection algorithm [31] is used to extract critical feature points from the edges; it outputs discrete key points as corners/features. This method improves upon the classic Harris corner detector by selecting points with strong gradients in orthogonal directions, ensuring reliable feature extraction.

Overall, by applying Laplacian-of-Gaussian edge detection followed by Shi–Tomasi point extraction on a binary sketch, we generate a clean set of connected contours and reproducible feature points. This process and another intermediary step that we will mention later on in this chapter, will then be used to form the nodes and edges of a rigid formation graph. The resulting graph representation provides a reliable, low-dimensional abstract of the user's drawing, ready for the assignment and control algorithms to faithfully reproduce the intended shape with a team of robots.

3.2 Linear Assignment: Hungarian Algorithm

The Hungarian algorithm, also known as the Kuhn-Munkres algorithm, solves the linear assignment problem in polynomial time by finding the optimal assignment that minimizes the total cost [34, 35]. This algorithm operates on a cost matrix where each row represents a task and each column represents a resource. The algorithm ensures that each task is assigned to exactly one resource in such a way that the total cost is minimized.

Given a cost matrix $C \in \mathbb{R}^{n \times n}$, where C_{ij} represents the cost of assigning robot j to formation node i, and n is the number of robots, the goal is to find an assignment such that the total cost, which is the sum of the costs C_{ij}, being the Euclidian distance between i and j, for the assigned pairs (i, j), is minimized. The optimization problem can be formulated as follows:

$$\min_{\mathbf{x}} \sum_{i=1}^{n} \sum_{j=1}^{n} C_{ij} x_{ij} \tag{3.1}$$

subject to the following constraints:

- *Assignment constraints:* Each robot is assigned to exactly one formation node:

$$\sum_{i=1}^{n} x_{ij} = 1 \quad \forall j \in \{1, 2, \ldots, n\} \tag{3.2}$$

- *Binary variables:* The decision variable x_{ij} is binary (either 0 or 1), where:

$$x_{ij} \in \{0, 1\} \quad \forall i, j \in \{1, 2, \ldots, n\} \tag{3.3}$$

In this work, our goal is to assign robots to formation nodes without introducing undesirable flips (i.e., maintaining the structural integrity of the formation). The Hungarian algorithm guarantees an optimal and collision-free assignment by selecting the best permutation of robots to nodes based on a cost function defined by the Euclidean distances between current and target nodes. By implementing the Hungarian algorithm, the assignment step ensures that rigid graph constraints are maintained while allowing translation and rotation of the formation but strictly avoiding flips.

3.3 Algorithm and Performance Guarantees

The flow of the proposed no-flip formation control algorithm is shown in Figure 3.3 and the details of its operations in Algorithm 1. As mentioned above, we used computer vision as a tool to translate human-drawn sketches to a rigid graph. The first step is to convert the sketch to a gray-scale image. Next, the gray-scale image serves as the input to an edge detector (here, the LoG edge detector) followed by a corner detector (here, the Shi-Tomasi corner detector). Having identified the most critical points describing the initial sketch sufficiently, scaling them into the space provided for the robots to be deployed in is a necessary next step. After constraining the points into a specific space, it is essential to ensure that a minimum distance separates the points in that space (especially when scaling down). We now have the set of points needed for the robots to construct the initial user drawing. The Laplacian matrix can then be computed to ensure a rigid graph construction. Subsequently, the most crucial part of guaranteeing a no-flip in the drawing involves the linear assignment of robots to appropriate formation nodes through the Hungarian algorithm. Finally, the formation control algorithm can be run to let the robots form the user-drawn sketch, using necessary and critical points.

Algorithm 1 HSI No-“Flip” Formation Control Algorithm

Require: User-drawn sketch image I
Ensure: Robot formation matching the sketch without flips
$I_{\text{gray}} \leftarrow$ Convert To Grayscale(I)
$E \leftarrow$ LoG Edge Detection(I_{gray})
$P \leftarrow$ ShiTomasi Corner Detection(E)
$P_{\text{scaled}} \leftarrow$ Scale Points(P)
$P_{\text{filt}} \leftarrow$ Filter Min Distance($P_{\text{scaled}}, d_{\text{min}}$)
Construct rigid graph $G = (V, E)$ from P_{filt}
$L \leftarrow$ Compute Laplacian(G)
$A \leftarrow$ Hungarian Assignment(L)
Apply formation-control law using L and assignment A
return Final robot positions in formation

From this algorithm, a no-flip guarantee theorem can be extracted. Let N be the number

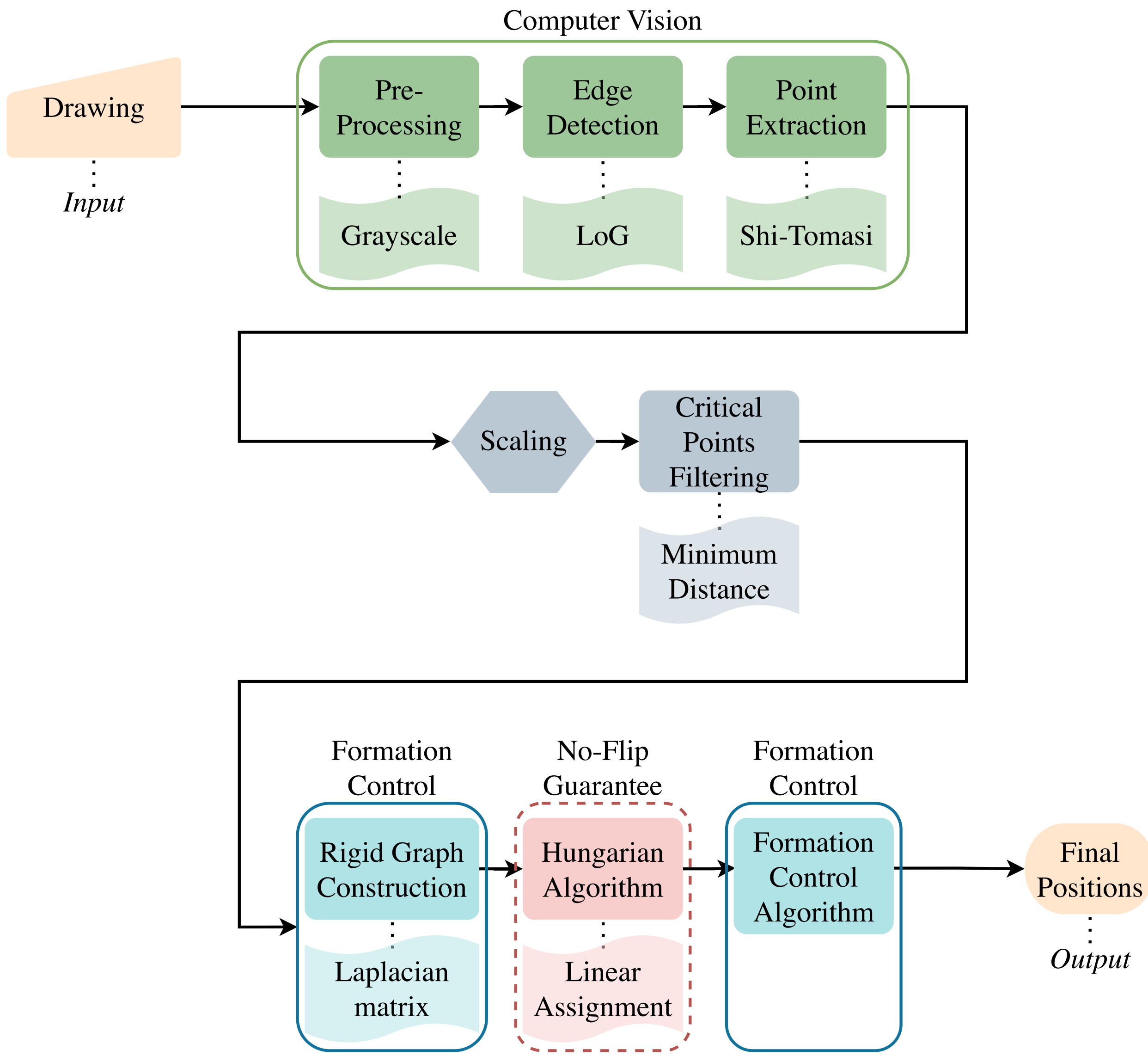


Figure 3.3: No-flip formation control algorithm.

of robots and $\mathcal{N} = \{1, \ldots, N\}$. $\mathcal{P}$ will be the set of all possible permutations over $\mathcal{N}$. We define the optimal permutation σ that maps each robot to its assigned formation node as follows:

$$\begin{aligned} &\sigma : i \to j, \quad i, j \in \{1, \ldots, n\}, \\ &\sigma(i) = j \quad \text{if and only if} \quad x_{ij} = 1. \end{aligned} \tag{3.4}$$

The angle between 3 points A, B, and C will be:

- defined as positive if the rotation from C to A around B is counterclockwise and negative otherwise, and
- the smallest of the two angles formed between segments BC and BA.

Let $\theta^{i,j,k}_{\text{agent}}(\sigma, p)$ be the angle between robots $\sigma(i)$, $\sigma(j)$, and $\sigma(k)$, where $i, j, k \in V$ and $\theta^{i,j,k}_{\text{target}}$ be the angle between formation nodes i, j, and k. Let $\phi(\sigma, p)$ be the total cost of the optimal permutation σ with respect to the robot initial positions p, defined as:

$$\phi(\sigma, p) = \sum_{i=1}^{n} \|p_{\sigma(i)} - q_i\|^2, \tag{3.5}$$

with q_i being the position of formation node i. Given this total cost function, we yield the following theorem.

Theorem 3.1. *If $\phi(\sigma, p) \leq \phi(\sigma', p)$ for all $\sigma' \in \mathcal{P}$, then*

$$sign[\theta^{i,j,k}_{agent}(\sigma, p)] = sign[\theta^{i,j,k}_{target}] \tag{3.6}$$

Proof. To prove by contradiction, assume that

$$\text{sign}[\theta^{i,j,k}_{\text{agent}}(\sigma, p)] \neq \text{sign}[\theta^{i,j,k}_{\text{target}}] \tag{3.7}$$

This means that the sign of the angle between robots $\sigma(i)$, $\sigma(j)$, and $\sigma(k)$ is opposite to that of the angle between formation nodes i, j, and k.

Using the Pythagorean theorem $a^2 = b^2 + c^2$ with a being the hypotenuse of the triangle, and b and c being the other two segments of the triangle, we get:

$$\|p_{\sigma(k)} - q_k\|^2 = \|p_{\sigma(k)} - q_i\|^2 + \|q_i - q_k\|^2 \tag{3.8}$$

$$\|p_{\sigma(i)} - q_i\|^2 = \|p_{\sigma(i)} - q_k\|^2 + \|q_k - q_i\|^2 \tag{3.9}$$

Adding those two equalities together, we get:

$$\begin{aligned}\|p_{\sigma(k)} - q_k\|^2 + \|p_{\sigma(i)} - q_i\|^2 = \\ \|p_{\sigma(k)} - q_i\|^2 + \|q_i - q_k\|^2 + \\ \|p_{\sigma(i)} - q_k\|^2 + \|q_k - q_i\|^2\end{aligned}$$

Rearranging, we get:

$$\begin{aligned}\|p_{\sigma(k)} - q_k\|^2 + \|p_{\sigma(i)} - q_i\|^2 = \\ \|p_{\sigma(k)} - q_i\|^2 + \|p_{\sigma(i)} - q_k\|^2 + \\ 2\|q_k - q_i\|^2\end{aligned}$$

Since $2\|q_k - q_i\|^2 \geq 0$, we get:

$$\begin{aligned}\|p_{\sigma(k)} - q_k\|^2 + \|p_{\sigma(i)} - q_i\|^2 \geq \\ \|p_{\sigma(k)} - q_i\|^2 + \|p_{\sigma(i)} - q_k\|^2\end{aligned}$$

Thus, the cost of the current assignment σ is not the minimum cost, thereby a flip, as defined in Definition 3.1, of robots i and k must occur. Hence, a contradiction is found, which proves our theorem. Figure 3.4 supports and helps visualize the theorem proof. □

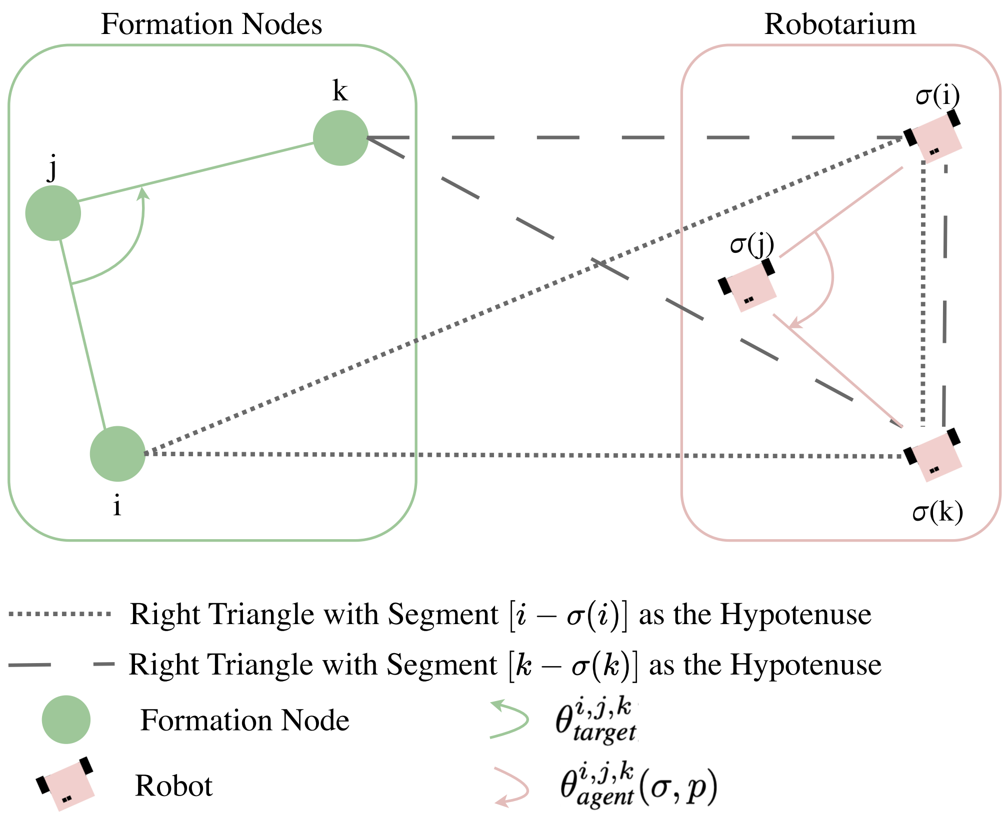


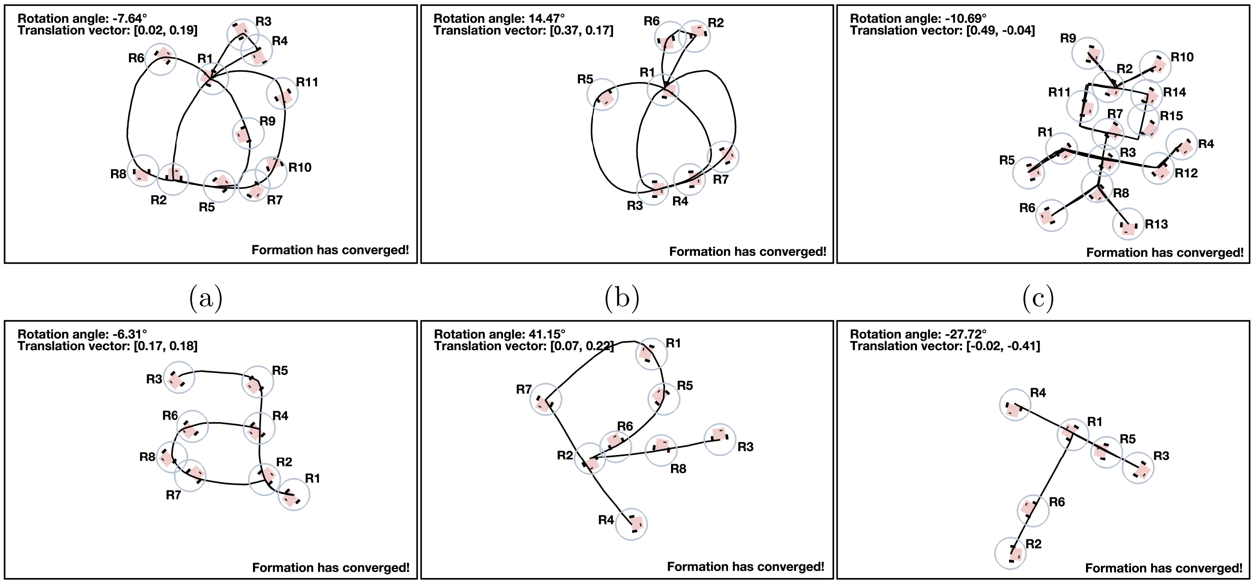
Figure 3.4: Visualization of the theorem contradiction proof where the assignments of robots cross paths and corresponding angles have opposite signs.

Chapter 4

Experimental Results

To validate the proposed formation control algorithm, we conducted a series of simulations using MATLAB and the Robotarium simulator as shown in Figure 4.1, as well as physical experiments in the UCI Robot Ecology Lab's in-house Robotarium as shown in Figures 4.2 and 4.3. The Robotarium allows for deployment of up to seven physical robots, necessitating an additional constraint in the deployment phase to ensure the algorithm functions effectively within the available robot number limit. As expected, with a smaller number of robots, accurately representing user-drawn sketches becomes more challenging. This can be shown in figure 4.1 (a) and (b) as the more robots are available, the more they can cover the needed points to represent a certain figure and form it accordingly.

In our proposed algorithm, as mentioned above, there is a no-flip guarantee; however, nothing constricts the robots to a certain strict final formation position. This leads to potential rotations and translations; we display those values in each of the simulations and the physical experiments for reference. Those rotations and translations are calculated by taking the initial scaled and filtered critical points in Step 5 of Algorithm 1 and comparing it with the final robots positions, which refer to the output of the algorithm (Step 10).



(a) (b) (c)

(d) (e) (f)

Figure 4.1: Simulation of robots representing non-flipped user-drawn: (a) pumpkin (unconstrained), (b) pumpkin (7 robots), (c) robot (unconstrained), (d) letter "a" (unconstrained), (e) letter "R" (unconstrained), and (f) letter "T" (unconstrained).

For simulation and testing purposes, these metrics can be calculated with the following optimization function:

$$F = \min_{\theta,\tau} \frac{1}{2} \sum_{i=1}^{N} \|R(\theta)p_i + \tau - q_i\|^2 \tag{4.1}$$

with:

- θ: Rotation angle aligning the initial drawing with the final robot positions.
- τ: Translation vector aligning the initial drawing with the final robot positions.
- N: Number of robots, corresponding to the number of initial points selected through the computer vision algorithm.
- p_i: Positions of the initial points before transformation.
- q_i: Final positions of the robots after executing the formation control algorithm.

We get that

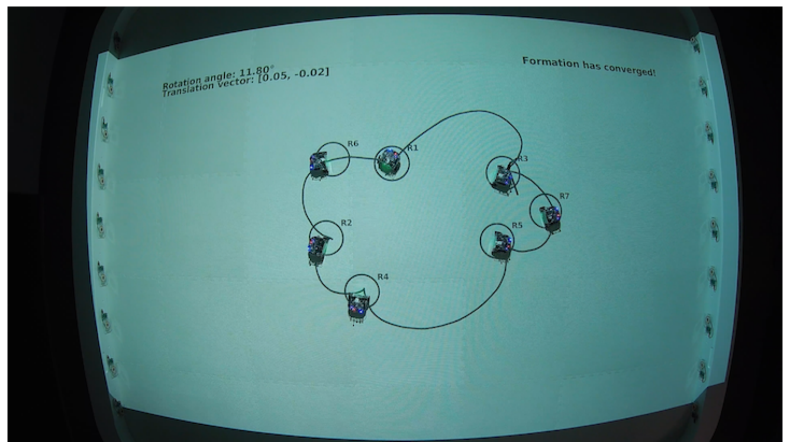
$$\frac{\partial F}{\partial \theta} = \sum_{i=1}^{N} [\tau^T R'(\theta) p_i - q_i^T R'(\theta) p_i] \tag{4.2}$$

and

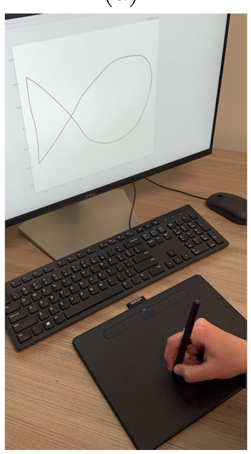
$$\frac{\partial F}{\partial \tau} = \sum_{i=1}^{N} [R(\theta) p_i - q_i + \tau] \tag{4.3}$$

and use gradient descent to reach the optimal rotation angle in degrees, and translation 2D vector in meters. More details on the derivation of Equations (4.2) and (4.3) can be found in Appendix A.

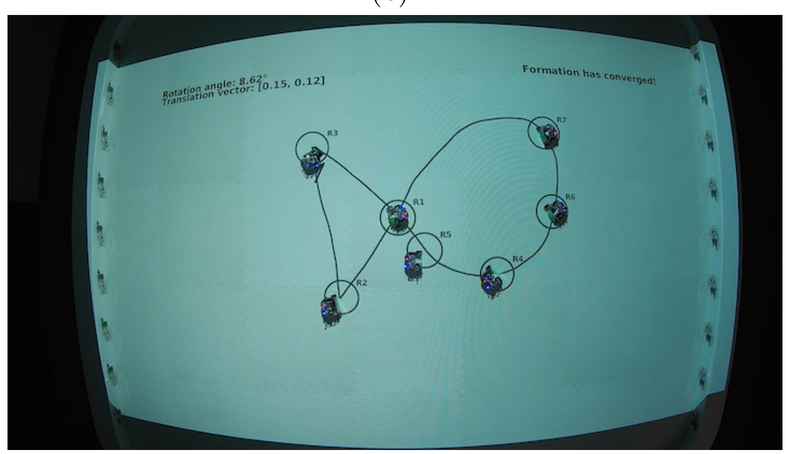
(a)



(b)

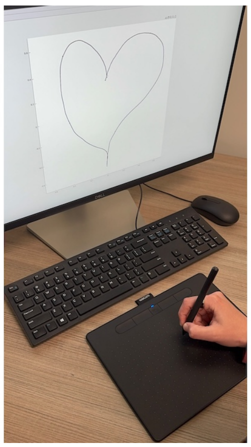


(c)

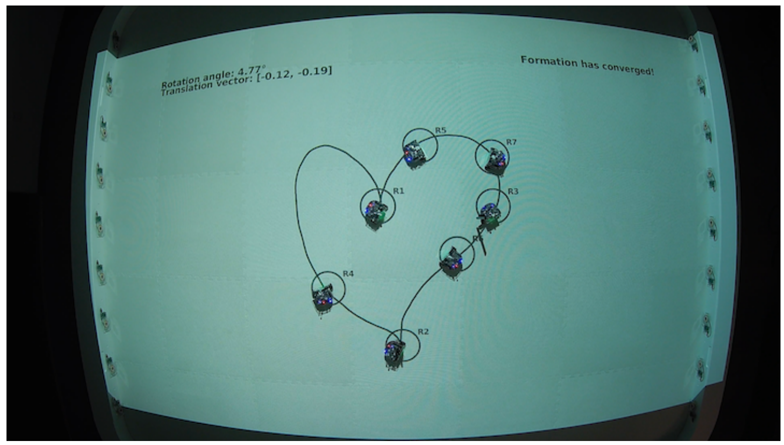


(d)

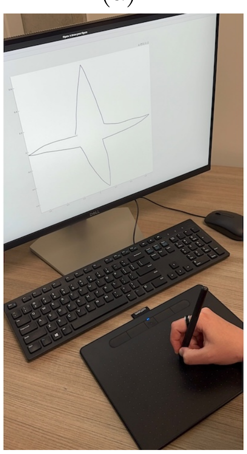
Figure 4.2: User drawing shapes on the graphical interface using a wireless tablet and the 7 robots in the robotarium representing the non-flipped drawings: (a, b) cloud and (c, d) fish.



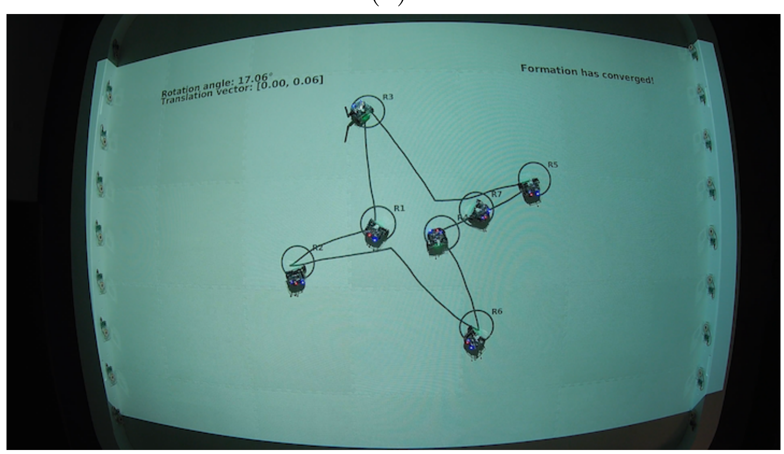
(a)



(b)



(c)



(d)

Figure 4.3: User drawing shapes on the graphical interface using a wireless tablet and the 7 robots in the robotarium representing the non-flipped drawings: (a, b) heart and (c, d) star.

Chapter 5

Conclusions and Future Work

In this thesis, we introduced a novel approach to human-swarm interaction through a simple, and user-friendly interface designed specifically for the deployment and control of multi-robot formations. By enabling non-expert users to specify robot swarm configurations via hand-drawn sketches, our method presents broader accessibility and interactive flexibility, bridging the gap between human intention and robotic swarm behavior. Our proposed methodology allowed users to dynamically specify formations in real-time, without any pre-defined mathematical configurations on the formation.

Our approach integrates computer vision techniques for feature extraction, graph-theoretic modeling for spatial relationships, and the Hungarian algorithm for optimal robot-node assignment. Through mathematical analysis and proof, we provided theoretical guarantees regarding the structural integrity of generated formations, explicitly guaranteeing the avoidance of undesirable formation flips, thus distorting the shape of the initial user-drawn sketch.

Experimental and simulation results demonstrated the practical robustness, accuracy, and effectiveness of our approach, confirming its suitability for scalable and complex multi-robot scenarios. Overall, our method represents a significant step forward in human-swarm interaction, empowering users to intuitively and reliably control robotic swarms, thereby opening numerous avenues for future research and real-world applications.

Future work may explore extensions to three-dimensional formations, incorporating additional sensory feedback and real-time environmental adaptation. Investigating advanced machine learning techniques to enhance feature extraction accuracy and user experience, as well as developing collaborative interfaces to facilitate simultaneous multi-user control, present additional promising research directions.

Bibliography

[1] A. Kolling, P. Walker, N. Chakraborty, K. Sycara, and M. Lewis, "Human interaction with robot swarms: A survey," *IEEE Transactions on Human-Machine Systems*, vol. 46, no. 1, pp. 9–26, 2015.

[2] M. Divband Soorati, J. Clark, J. Ghofrani, D. Tarapore, and S. D. Ramchurn, "Designing a user-centered interaction interface for human–swarm teaming," *Drones*, vol. 5, no. 4, p. 131, 2021.

[3] S. O. Sachidanandam, S. Honarvar, and Y. Diaz-Mercado, "Effectiveness of augmented reality for human swarm interactions," in *Proceedings of the International Conference on Robotics and Automation (ICRA)*. IEEE, 2022, pp. 11 258–11 264.

[4] V. Serpiva, E. Karmanova, A. Fedoseev, S. Perminov, and D. Tsetserukou, "Swarmpaint: Human-swarm interaction for trajectory generation and formation control by dnn-based gesture interface," in *Proceedings of the International Conference on Unmanned Aircraft Systems (ICUAS)*. IEEE, 2021, pp. 1055–1062.

[5] M. Macchini, L. De Matteïs, F. Schiano, and D. Floreano, "Personalized human-swarm interaction through hand motion," *IEEE Robotics and Automation Letters*, vol. 6, no. 4, pp. 8341–8348, 2021.

[6] D. S. Drew, "Multi-agent systems for search and rescue applications," *Current Robotics Reports*, vol. 2, pp. 189–200, 2021.

[7] L. Wang, D. Zhu, W. Pang, and Y. Zhang, "A survey of underwater search for multi-target using multi-AUV: Task allocation, path planning, and formation control," *Ocean Engineering*, vol. 278, p. 114393, 2023.

[8] C. Lytridis, V. G. Kaburlasos, T. Pachidis, M. Manios, E. Vrochidou, T. Kalampokas, and S. Chatzistamatis, "An overview of cooperative robotics in agriculture," *Agronomy*, vol. 11, no. 9, 2021.

[9] J. Huang, G. Tian, J. Zhang, and Y. Chen, "On unmanned aerial vehicles light show systems: Algorithms, software and hardware," *Applied Sciences*, vol. 11, no. 16, 2021.

[10] B. Andrievsky, A. M. Popov, I. Kostin, and J. Fadeeva, "Modeling and control of satellite formations: A survey," *Automation*, vol. 3, no. 3, pp. 511–544, 2022.

[11] R. Olfati-Saber, J. A. Fax, and R. M. Murray, "Consensus and cooperation in networked multi-agent systems," *Proceedings of the IEEE*, vol. 95, no. 1, pp. 215–233, 2007.

[12] M. Ji and M. Egerstedt, "Distributed coordination control of multiagent systems while preserving connectedness," *IEEE Transactions on Robotics*, vol. 23, no. 4, pp. 693–703, 2007.

[13] R. Olfati-Saber and R. M. Murray, "Graph rigidity and distributed formation stabilization of multi-vehicle systems," in *Proceedings of the 41st IEEE Conference on Decision and Control*, vol. 3. IEEE, 2002, pp. 2965–2971.

[14] J. A. Fax and R. M. Murray, "Information flow and cooperative control of vehicle formations," *IEEE Transactions on Automatic Control*, vol. 49, no. 9, pp. 1465–1476, 2004.

[15] A. Suresh and S. Martínez, "Human-swarm interactions for formation control using interpreters," *International Journal of Control, Automation and Systems*, vol. 18, no. 8, pp. 2131–2144, 2020.

[16] Y. Cao, W. Yu, W. Ren, and G. Chen, "An overview of recent progress in the study of distributed multi-agent coordination," *IEEE Transactions on Industrial informatics*, vol. 9, no. 1, pp. 427–438, 2012.

[17] M. A. Lewis and K.-H. Tan, "High precision formation control of mobile robots using virtual structures," *Autonomous robots*, vol. 4, pp. 387–403, 1997.

[18] K.-H. Tan and M. A. Lewis, "Virtual structures for high-precision cooperative mobile robotic control," in *Proceedings of IEEE/RSJ International Conference on Intelligent Robots and Systems. IROS*, vol. 1. IEEE, 1996, pp. 132–139.

[19] R. Olfati-Saber, "Flocking for multi-agent dynamic systems: Algorithms and theory," *IEEE Transactions on Automatic Control*, vol. 51, no. 3, pp. 401–420, 2006.

[20] W. Ren and R. W. Beard, "Consensus seeking in multiagent systems under dynamically changing interaction topologies," *IEEE Transactions on Automatic Control*, vol. 50, no. 5, pp. 655–661, 2005.

[21] M.-C. Park, Z. Sun, B. D. Anderson, and H.-S. Ahn, "Distance-based control of kn formations in general space with almost global convergence," *IEEE Transactions on Automatic Control*, vol. 63, no. 8, pp. 2678–2685, 2017.

[22] G. Laman, "On graphs and rigidity of plane skeletal structures," *Journal of Engineering Mathematics*, vol. 4, no. 4, pp. 331–340, 1970.

[23] W. Yu, G. Chen, and M. Cao, "Some necessary and sufficient conditions for second-order consensus in multi-agent dynamical systems," *Automatica*, vol. 46, no. 6, pp. 1089–1095, 2010.

[24] M. Mesbahi and M. Egerstedt, “Graph theoretic methods in multiagent networks,” 2010.

[25] A. Rasouli and J. K. Tsotsos, “Autonomous vehicles that interact with pedestrians: A survey of theory and practice,” *IEEE Transactions on Intelligent Transportation Systems*, vol. 21, no. 3, pp. 900–918, 2019.

[26] R. Szeliski, *Computer vision: algorithms and applications.* Springer Nature, 2022.

[27] R. C. Gonzales and P. Wintz, *Digital image processing.* Addison-Wesley Longman Publishing Co., Inc., 1987.

[28] J. Canny, “A computational approach to edge detection,” *IEEE Transactions on Pattern Analysis and Machine Intelligence*, no. 6, pp. 679–698, 1986.

[29] D. Marr and E. Hildreth, “Theory of edge detection,” *Proceedings of the Royal Society of London. Series B. Biological Sciences*, vol. 207, no. 1167, pp. 187–217, 1980.

[30] C. Harris and M. Stephens, “A combined corner and edge detector,” in *Proceedings of the 4th Alvey Vision Conference*, 1988, pp. 147–151.

[31] J. Shi *et al.*, “Good features to track,” in *1994 Proceedings of IEEE Conference on Computer Vision and Pattern Recognition.* IEEE, 1994, pp. 593–600.

[32] E. Rosten and T. Drummond, “Machine learning for high-speed corner detection,” in *Computer Vision–ECCV 2006: 9th European Conference on Computer Vision, Graz, Austria, May 7-13, 2006. Proceedings, Part I 9.* Springer, 2006, pp. 430–443.

[33] D. G. Lowe, “Distinctive image features from scale-invariant keypoints,” *International Journal of Computer Vision*, vol. 60, pp. 91–110, 2004.

[34] H. W. Kuhn, “The hungarian method for the assignment problem,” *Naval Research Logistics Quarterly*, vol. 2, no. 1-2, pp. 83–97, 1955.

[35] J. Munkres, “Algorithms for the assignment and transportation problems,” *Journal of the Society for Industrial and Applied Mathematics*, vol. 5, no. 1, pp. 32–38, 1957.

Appendix A

Derivation of the Optimal Translation and the Rotation Gradient

We wish to solve

$$\min_{\theta,\tau} F(\theta,\tau) = \min_{\theta,\tau} \frac{1}{2}\sum_{i=1}^{N} \|R(\theta)p_i + \tau - q_i\|^2 \tag{A.1}$$

for the translation $\tau \in \mathbb{R}^2$ and rotation angle $\theta \in \mathbb{R}$, with:

- θ: Rotation angle aligning the initial drawing with the final robot positions.
- τ: Translation vector aligning the initial drawing with the final robot positions.
- N: Number of robots, corresponding to the number of initial points selected through the computer vision algorithm.
- p_i: Positions of the initial points before transformation.
- q_i: Final positions of the robots after executing the formation control algorithm.

Here

$$R(\theta) = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}, \quad R'(\theta) = \frac{dR}{d\theta} = \begin{pmatrix} -\sin\theta & -\cos\theta \\ \cos\theta & -\sin\theta \end{pmatrix}. \tag{A.2}$$

1. Closed-form for τ. Expand the quadratic in τ:

$$\begin{aligned} F(\theta,\tau) &= \frac{1}{2}\sum_{i=1}^{N}\Big[\big(R(\theta)p_i + \tau - q_i\big)^T\big(R(\theta)p_i + \tau - q_i\big)\Big] && \text{(A.3)} \\ &= \frac{1}{2}\sum_{i=1}^{N}\Big[\big(\tau + (R(\theta)p_i - q_i)\big)^T\big(\tau + (R(\theta)p_i - q_i)\big)\Big] && \text{(A.4)} \end{aligned}$$

Differentiating w.r.t. τ and setting to zero gives

$$\frac{\partial F}{\partial \tau} = \sum_{i=1}^{N}\big[\tau + \big(R(\theta)p - q\big)\big] = N\big[\tau + \big(R(\theta)p - q\big)\big] \implies \tau^\star = q \;-\; R(\theta)\,p. \tag{A.5}$$

2. Closed-form for θ. Expand the quadratic in θ:

$$\begin{aligned} F(\theta,\tau) &= \frac{1}{2}\sum_{i=1}^{N}\Big[\big(R(\theta)p_i + \tau - q_i\big)^T\big(R(\theta)p_i + \tau - q_i\big)\Big] && \text{(A.6)} \\ &= \frac{1}{2}\sum_{i=1}^{N}\Big[\big(p_i^T R(\theta)^T + \tau^T - q_i^T\big)\big(R(\theta)p_i + \tau - q_i\big)\Big] && \text{(A.7)} \\ &= \frac{1}{2}\sum_{i=1}^{N}\Big[2p_i^T R^T(\theta)\tau - 2p_i^T R^T(\theta)q_i + \dots\Big] && \text{(A.8)} \end{aligned}$$

We omit the rest of the factors since they do not contain the variable θ thus not affecting the rest of the work. Differentiating w.r.t. θ gives

$$\frac{\partial F}{\partial \theta} = \sum_{i=1}^{N}\Big[\tau^T R'(\theta)p_i - q_i^T R'(\theta)p_i\Big] = N\Big[\tau^T R'(\theta)p - q^T R'(\theta)p\Big]. \tag{A.9}$$

3. Gradient-descent updates. We then iterate

$$\theta_{k+1} = \theta_k \; - \; \alpha_{small} \frac{\partial F}{\partial \theta}\Big|_{\theta_k}, \qquad \tau_{k+1} = q - R(\theta_{k+1})\, p, \tag{A.10}$$

until convergence.

This completes the derivation of the closed-form translation and the rotation gradient that can be used in the gradient-descent scheme.